UNIVERSITY OF CAMBRIDGE

Department of Computer Science and Technology

Prof. Nicholas D. Lane
Professor of Machine Learning Systems
Fellow of St John's College
Co-founder & CSO, Flower Labs

13th March, 2025

To whom it may concern,

My name is Nicholas Lane, lead instructor of the course 'Deep Learning' and I certify that student Li Enxin completed the course over 2 weeks of study in Cambridge, UK during February 2025.
Please note the course was run by Oxbridge Neural Group and is not part of the University of Cambridge in any way.

The course provided an introduction to Deep Learning and introduced core competencies of the subject in an easy to understand and accessible manner that didn't require any previous knowledge of or experience with programming, statistics, or deep learning. Students developed a fundamental knowledge and the skills necessary for further study of deep learning.

The student swiftly mastered essential concepts and dedicated considerable effort to align with the program's pace. Demonstrating excellent reasoning skills, they posed insightful questions, deepening their understanding of key concepts. Active participation in small group discussions and exercises showcased their ability to collaborate seamlessly with peers, and they confidently led discussions at times.

I am confident that Li would thrive in advanced studies of the topic.

Best,





Nic Lane — Oxbridge Neural Group | University of Cambridge | Flower Labs

Filip Svoboda — Oxbridge Neural Group | University of Cambridge

Bio: Nic Lane is a full Professor in the department of Computer Science and Technology at the University of Cambridge, where he leads the Machine Learning Systems lab (CaMLSys). Nic is also a Fellow of St. John's College. Alongside his academic appointments, he is the co-founder and Chief Scientific Officer of Flower Labs (https://flower.dev/), a venture-backed AI company (YCW23) behind the Flower framework. Until early this year, Nic was the Lab Director at Samsung AI in Cambridge. He was responsible for the technical direction of this 50-person lab that has an agenda of broadly advancing artificial intelligence in service of future Samsung products and services; and in addition to leading the lab -- Nic personally directed teams focused on distributed and on-device forms of learning. Prior to his role at the University of Cambridge, Nic was an Associate Professor in the Computer Science department at the University of Oxford. At Oxford, he was a member of the Cyber-Physical Systems group, taught machine learning on the Professional Masters Program (PMP) and was a Fellow of Kellogg College. Earlier in his career, Nic held dual appointments at University College London (UCL) and Nokia Bell Labs; at Nokia, as a Principal Scientist. In recent years, Nic has specialised in the study of efficient machine learning under computational constraints, and over the last seven years he has pioneered a range of embedded and mobile forms of deep learning. Nic has received multiple best paper awards, including ACM/IEEE IPSN 2017 and two from ACM UbiComp (2012 and 2015). In 2018 and 2019, he (and his co-authors) received the ACM SenSys Test-of-Time award and ACM SIGMOBILE Test-of-Time award for pioneering research, performed during his PhD thesis, that devised machine learning algorithms used today on devices like smartphones. Nic served as the PC-chair of ACM MobiSys 2019, a role he has performed also for ACM HotMobile and ACM SenSys in the past. Nic was the 2020 ACM SIGMOBILE Rockstar award winner for his contributions to "the understanding of how resource-constrained mobile devices can robustly understand, reason and react to complex user behaviours and environments through new paradigms in learning algorithms and system design." Prior to moving to England, Nic spent 4-years at Microsoft Research based in Beijing as a Lead Researcher. He received his PhD from Dartmouth College in 2011. Nic also received an M.Eng from Cornell University (2004) and a B.Sc (Hons) from the University of Waikato (2001)

Artificial Intelligence and Deep Learning

Course overview and mark breakdown

Lecture 1 (week 1)	Lecture 1 will introduce background material that explains what made Deep Learning the juggernaut it is today and the forces that are likely to shape its evolution in the future.	95%
Lecture 2 (week 1)	Lecture 2 will introduce the core building blocks that power deep models. Traditional architectures, convolutional and recurrent networks, will be covered. Applications in vision and natural language processing will be discussed and code examples will be presented.	97%
Lecture 3 (week 1)	Lecture 3 will introduce the advanced building blocks that power modern deep models. Residual connections, depth wise separability, channel shuffle, and attention will be covered. Applications in vision and natural language processing will be discussed and code examples presented.	93%
Lecture 4 (week 1)	Lecture 4 will examine the art of training deep neural networks. First a solid understanding of deep learning parameter learning will be derived from core statistical concepts. Then a step-by-step guide to successfully training a deep model will be built on top of this foundation.	76%
Lecture 5 (week 2)	Lecture 5 will explore the fundamentals of deep learning computational resource management. The chief impediment to further deep learning adoption is its dependence on expensive large-scale compute. This lecture will tackle this dependence both from the vantage point of computer systems as well as the algorithms themselves.	88%
Lecture 6 (week 2)	Lecture 6 will take a deep dive into the hottest topics in the bleeding-edge deep learning literature. The most exciting and influential deep learning papers will be presented, and their methods will be analysed in detail. The students will learn how to critically assess and learn from the best of the best papers. in Deep Learning.	100%
Lecture 7 (week 2)	Lecture 7 will be structured as a student-led reading group. The students will be given the opportunity to put the knowledge they learned over the preceding 5 weeks to test and practice their academic communication, presentation, and research evaluation skills by delivering concise summaries of the most important papers in deep learning.	85%
	Final Grade	**A**